



ENEIDE: a High-Quality Silver Standard Dataset for Named Entity Recognition and Linking in Historical Italian

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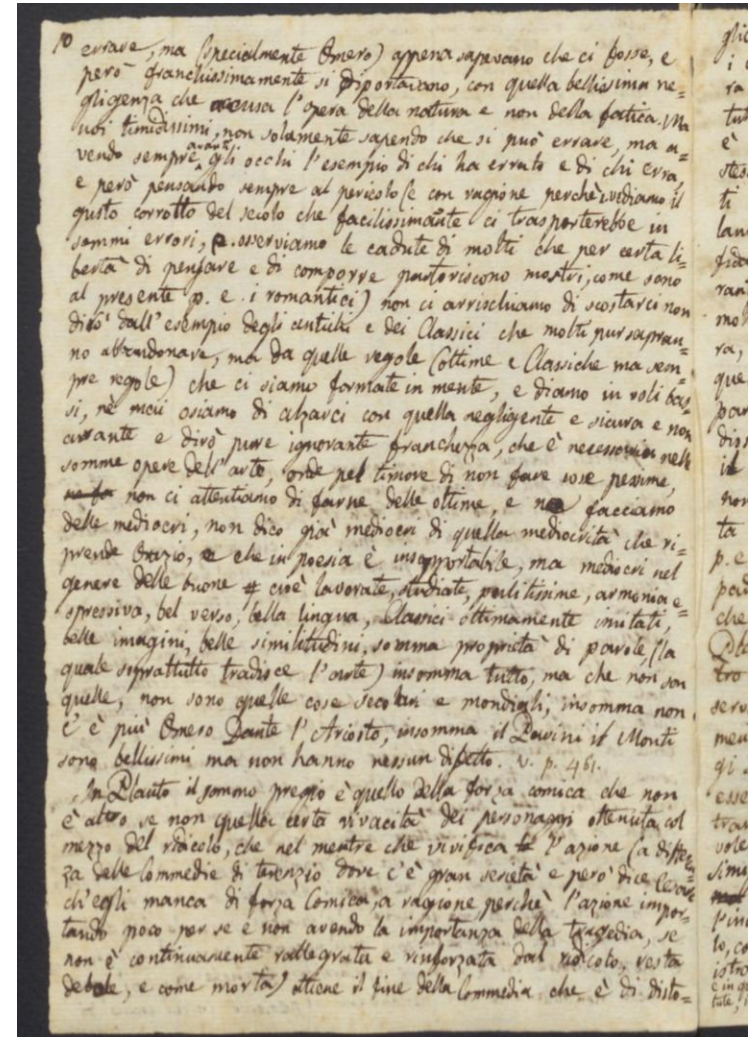
LREC 20
Palma **26**





Outline

- Why a Silver Standard?
- Italian SDEs: DigitalZibaldone & Aldo Moro Digitale
- Dataset Curation
- NER Results
- Entity Linking Results



Page of Leopardi's Zibaldone in the Napoli National Library



Named Entity Processing in the Humanities

The sayings, responses, etc., that [Machiavelli](#) attributes to [Castruccio Castracani](#) (in his *Life of him*) are all, or almost all, the very same ones that [Diogenes Laërtius](#) and others report about ancient philosophers, with only the names, places, etc., changed.

From Giacomo Leopardi, *Zibaldone di Pensieri*, p. 4368 par. 2 (1828)

Named Entity Recognition (NER):

- *Machiavelli, Castruccio Castracani* and *Diogenes Laërtius* are **Person**
- *Life of him* is a **Work**

Entity Linking (EL):

- *Machiavelli* → [Q1399](#)
- *Castruccio Castracani* → [Q1049990](#)
- *Diogenes Laërtius* → [Q59138](#)
- *Life of him* → [Q3824985](#)





Current Gaps in Historical Datasets

- Italian historical texts represent a vast, underexploited resource for research in NLP and DH (Ehrmann et al., 2023)
- Existing gold-standard datasets are expensive to create and limited in scope (Graciotti et al., 2025)
 - No existing dataset to train EL models on historical Italian
- Scholarly Digital Editions (SDEs) offer curated historical texts already annotated by domain experts, rich in named entities annotations

- Ehrmann, M. et al. (2023). Named Entity Recognition and Classification in Historical Documents: A Survey. *ACM Computing Surveys*.
- Graciotti, A. et al. (2025). KE-MHISTO: Towards a multilingual historical knowledge extraction benchmark for addressing the long-tail problem. *ACL*.



SDEs

« 4368,1

4369,1»



I detti, risposte ec. che [Machiavelli](#) attribuisce a [Castruccio Castracani](#) (nella [Vita di questo](#)), sono tutti o quasi tutti gli stessissimi che il [Laerzio](#) ec. riferiscono di filosofi antichi, mutati solo i nomi, i luoghi ec. [Machiavelli](#) del resto non sapeva il greco, poco o nulla il latino, ed era poco letterato. Non sarebbe maraviglia ch'egli avesse seguito una tradizione popolare che avesse conservati que' motti mutando i nomi, e attribuendoli al personaggio nazionale di [Castruccio](#), noto per singolare acutezza e prontezza d'ingegno. Il popolo fiorentino racconta ancora di [Dante](#) e dello stesso [Machiavello](#) vari tratti che si leggono negli antichi greci e latini, come quello di [Esopo](#) che diede un asse a chi gli tirò una sassata ec., il qual tratto (con modificazioni accidentali e non di sostanza) si racconta dal volgo in [Firenze](#) di [Machiavelli](#). (Tengo queste cose da [Forti](#) e da [Capei](#)). Così non solo le nazioni, ma le città, tirano alla storia ed a' personaggi propri, e in somma alle cose ed alle persone a se più cognite, i fatti delle storie altrui, noti al volgo per antiche tradizioni orali. A [Napoli](#) resta ancora in proverbio la sapienza e dottrina di [Abelardo](#): [4369](#) ne sa più di [Pietro Abailardo](#) ([Capei](#)). In ogni modo quel libro di [Machiavelli](#) farebbe sempre al mio proposito molto bene. V. p. 4430.

Passage contained in Giacomo Leopardi's Zibaldone – Digital Edition

URL: https://digitalzibaldone.net/node/p4368_2



Contributions

1. Present a novel NERL dataset in historical Italian called ENEIDE
(Extracting Named Entities from Italian Digital Editions)
2. Investigate how ENEIDE can be used to advance NER and EL techniques facing the diachronic challenges of Italian historical texts

DigitalZibaldone & Aldo Moro Digitale

- ENEIDE is extracted from two SDEs: **DigitalZibaldone** (DZ) (1817 - 1832) and **Aldo Moro Digitale** (AMD) (1936 - 1976)
 - Literary (DZ) and political (AMD) domains
- Contains manual annotations to named entities of type *person, location, literary work* and *organization* linked to Wikidata



- Silvia Stoyanova and Ben Johnston (Ed.), *Giacomo Leopardi's Zibaldone di pensieri: a digital research platform*. <https://digitalzibaldone.net/>
- Barzaghi, S. et al. (2025). The Semantic Digital Edition of Aldo Moro's Writings: A Workflow supporting Data Sharing and Replicability. *ACM Journal on Computing and Cultural Heritage*. <https://aldomorodigitale.unibo.it/>



Annotation Extraction

« 2718,1

2721,2»

[2721,1] Anche il [Gelli](#) confessava (ap. [Perticari Degli Scritt. del Trecento](#) I. 2. c. 13. p. 183.) che la lingua toscana non era stata applicata alle scienze. (24. Maggio 1823.).

https://digitalzibaldone.net/node/p2721_1

doc_id	text	publication_date
p2721_1	Anche il Gelli confessava (ap. Perticari [...])	1823-05-24



doc_id	surface	start_pos	end_pos	type	identifier
p2721_1	Gelli	9	14	PER	Q518160
p2721_1	Perticari	31	40	PER	Q3769747
p2721_1	Degli Scritt. del Trecento	41	67	WORK	NIL





Sampling and Assessment

- **DZ:** 1050 notes sampled from Leopardi's most productive years (1821 - 1823)
- **AMD:** 1061 initial paragraphs from each document (excluding outliers in length)
- Quality assessment:
 - 100 random documents from each edition
 - Extracted annotations compared to human annotations
- **Extended annotation coverage on AMD with dictionary-based methods and NER model**

Statistics	DZ	AMD
Annotations	403	215
Wrong annotations	13	18
Missing annotations	10	64
Precision	96.8	91.6
Recall	94.4	70.6
F1	95.6	79.8



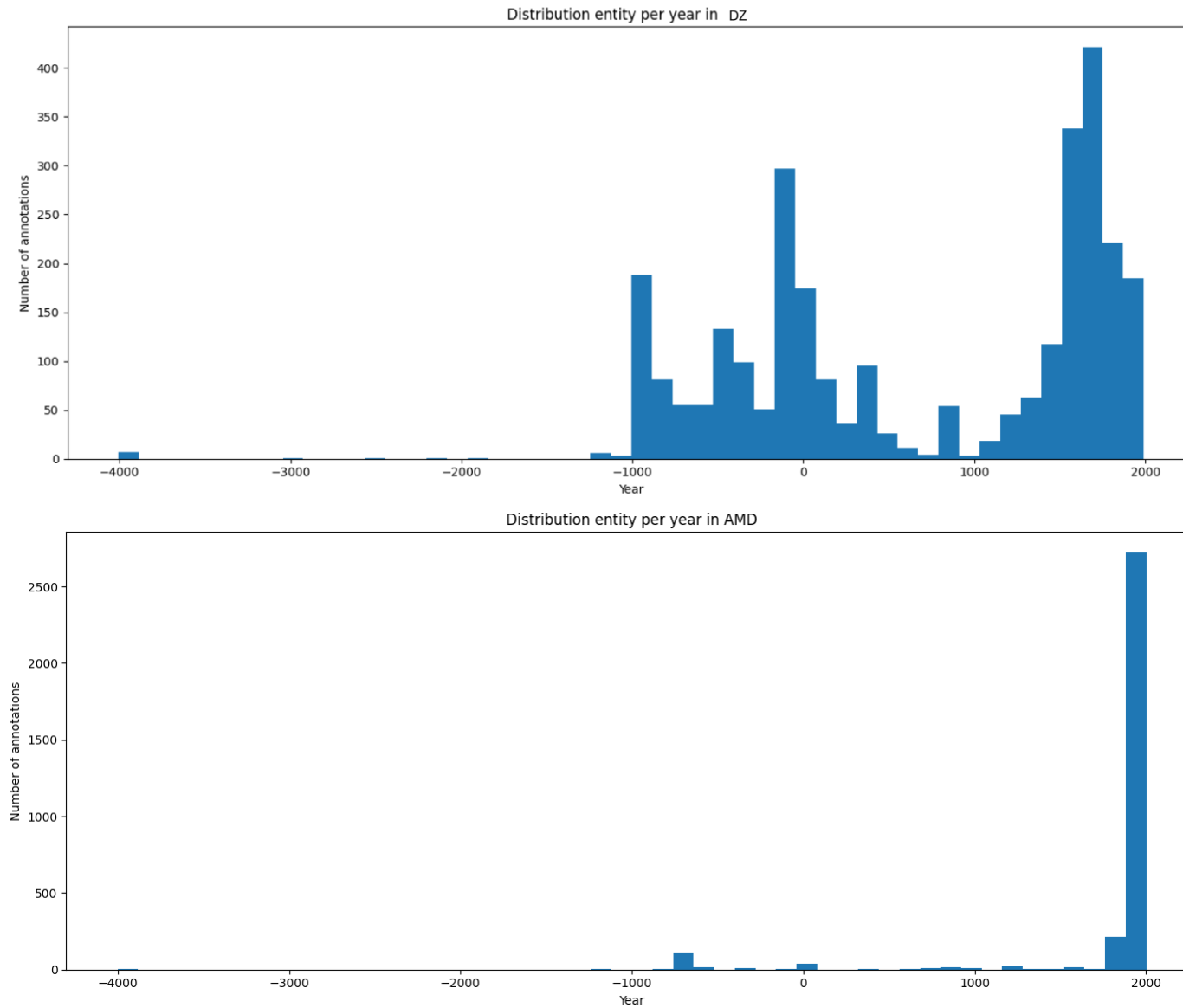
Dataset Statistics

- DZ has a higher number of NIL entities (298) compared to AMD (86)
- DZ has a higher overlap due to more temporal and thematic homogeneity

	DZ	AMD
Documents (train)	735	743
Documents (dev)	157	159
Documents (test)	158	160
Tokens per Doc (train)	246.6	125.9
Tokens per Doc (dev)	253.2	119.3
Tokens per Doc (test)	248.7	119
Annotations (train)	2,935	2,766
Annotations (dev)	727	604
Annotations (test)	617	657
% Identifiers Overlap (train+dev vs test)	93.19	75.38



Temporal Distribution of Entities in ENEIDE





Evaluation Setup

- **NER:**

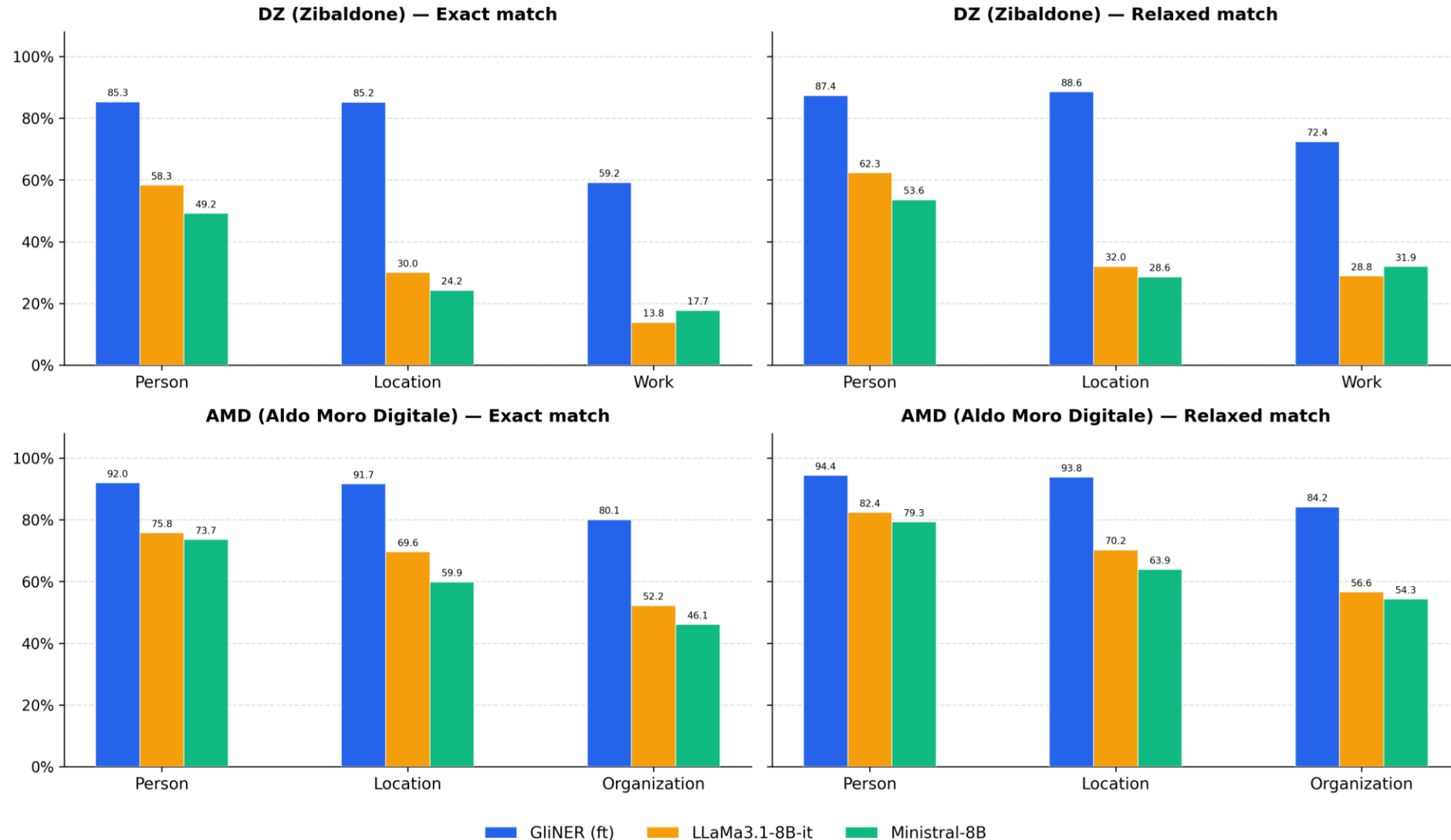
- Fine-tuned GliNER and 2 8B instruction-tuned LLMs (LLaMa3.1 and Ministral) prompted for zero-shot NER
- Visualization of F1 scores per-class (exact and *fuzzy*)

- **EL:**

- BLINK (Italian bi-encoder), mGENRE (auto-regressive), ChatEL (bi-encoder + Mistral-24B for candidate selection)
- Visualization of accuracy scores per-class

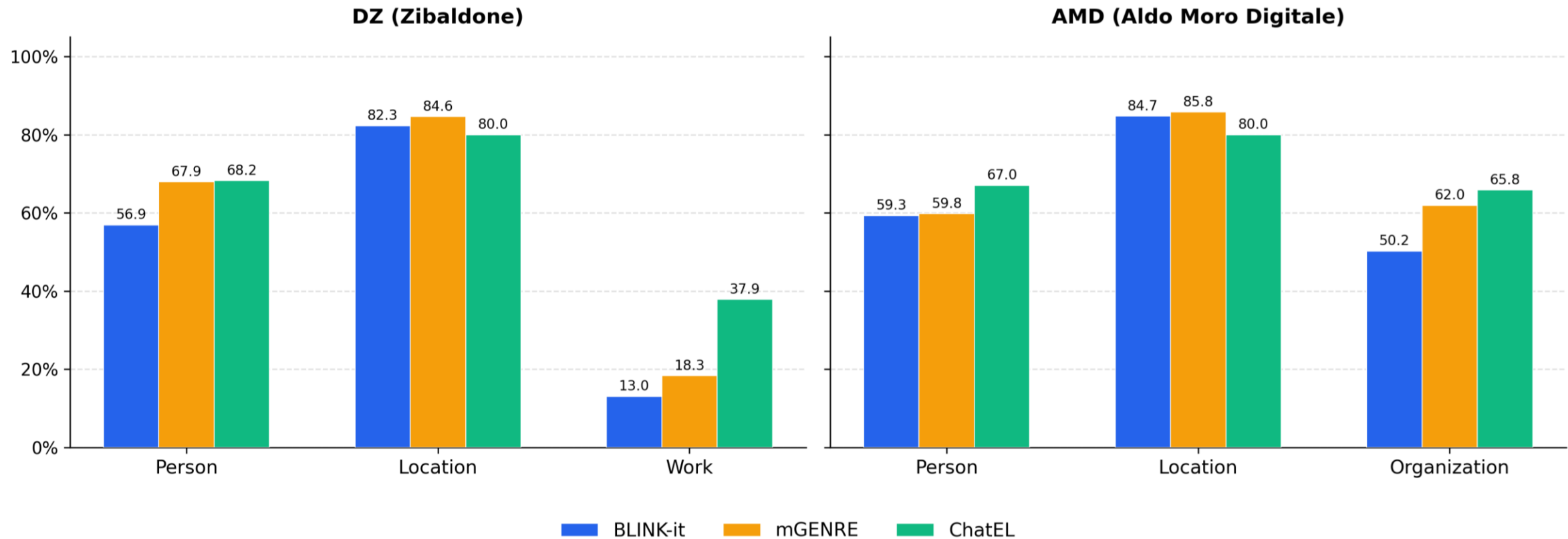


NER Results: Per-class F1





EL Results: Per-class Accuracy





Future Perspectives

- **Benchmarking:** Collect further annotated documents to benchmark NER and EL on Italian texts from different periods
- **NIL Clustering:** Extend the benchmark to cluster NIL entities
- **Model Improvement:** Develop methodologies for integrating contextual knowledge to refine NER and EL predictions





Thank you! Questions?



Dataset



Preprint

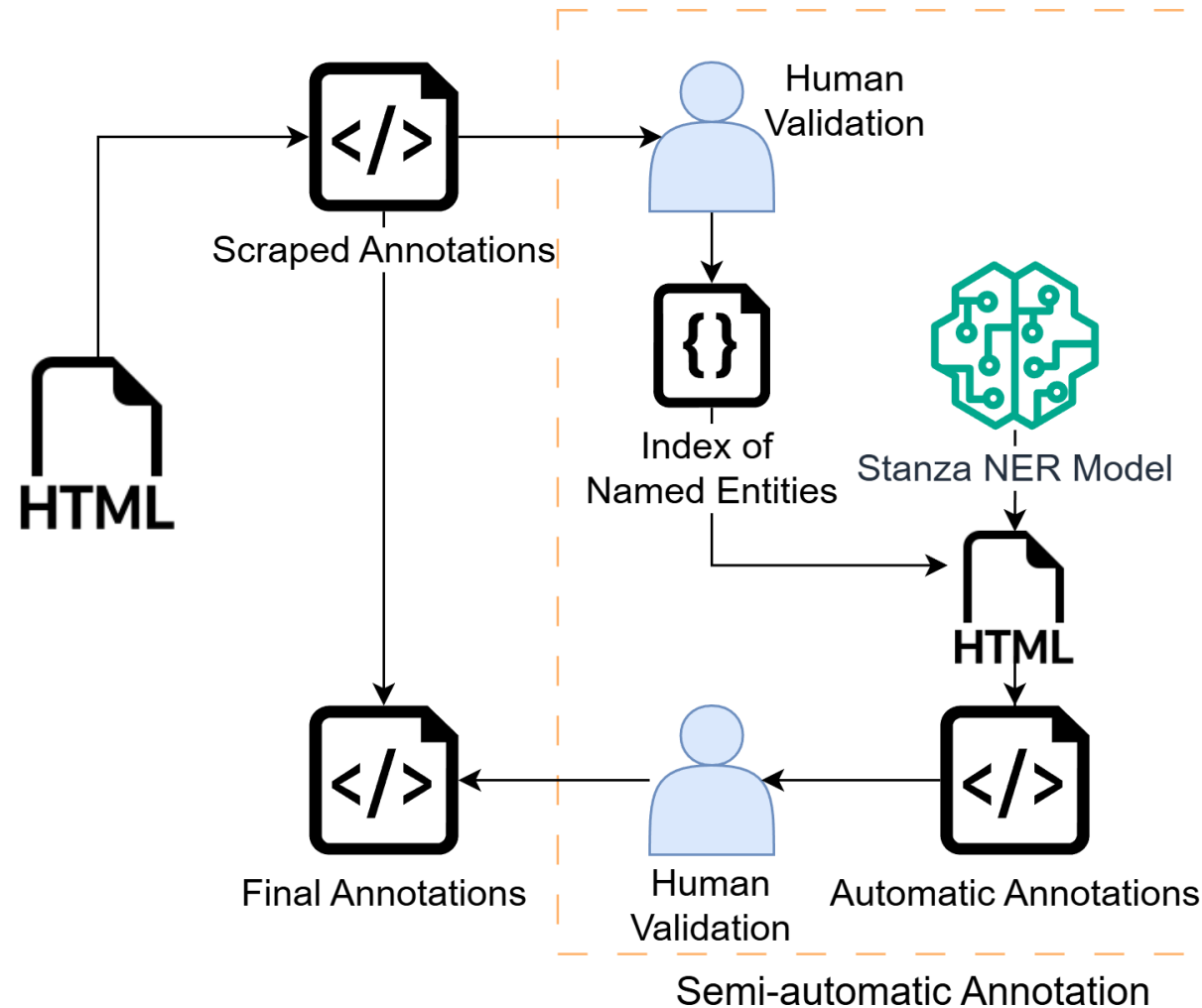
E-mail: c.santini12@unimc.it



Extra Material



Semi-automatic Enrichment for AMD





References

- Barzaghi, S. et al. (2025). The Semantic Digital Edition of Aldo Moro's Writings: A Workflow supporting Data Sharing and Replicability. *ACM Journal on Computing and Cultural Heritage*. <https://aldomorodigitale.unibo.it/>
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- Graciotti, A. et al. (2025). KE-MHISTO: Towards a multilingual historical knowledge extraction benchmark for addressing the long-tail problem. *ACL*.
- Pozzi, R. et al. (2023). Named entity recognition and linking for entity extraction from italian civil judgements. *AlxIA*.
- Silvia Stoyanova and Ben Johnston (Ed.), *Giacomo Leopardi's Zibaldone di pensieri: a digital research platform*. <https://digitalzibaldone.net/>
- Zaratiana, U. et al. (2024) "GLiNER: Generalist model for named entity recognition using bidirectional transformer." *NAACL*.